Class #13 - The Gram-Schmidt Process

1. Theorem: If $\{u_1, \cdots, u_p\}$ is an orthonormal basis for a subspace W of $\mathbf{R^n}$, then

$$\text{Proj}_W y = \underline{\qquad (y \cdot u_1) u_1 + \cdots + (y \cdot u_p) u_p \qquad}.$$

If $U = [u_1\ u_2\ \cdots\ u_p]$, then

$$\text{Proj}_W y = \underline{\qquad U U^T y \qquad} \quad \forall y \in \mathbf{R^n}.$$

Proof:

$\text{Proj}_W y$ = "orthogonal proj. of y onto W"

$= (y \cdot u_1) u_1 + \cdots + (y \cdot u_p) u_p$

$= \begin{bmatrix} | & & | \\ u_1 & \cdots & u_p \\ | & & | \end{bmatrix} \begin{bmatrix} y \cdot u_1 \\ \vdots \\ y \cdot u_p \end{bmatrix}$

$= U \begin{bmatrix} - u_1^T - \\ \vdots \\ - u_p^T - \end{bmatrix} \begin{bmatrix} | \\ y \\ | \end{bmatrix} = U U^T y$

2. The Gram-Schmidt process is an algorithm for producing an _orthogonal_ or _orthonormal_ basis for any _nonzero_ subspace of $\mathbf{R^n}$.

3. Example: Let $b_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $b_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $b_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

Then $\{b_1, b_2, b_3\}$ is a basis for a subspace W of $\underline{\qquad \mathbb{R}^4 \qquad}$. Construct an orthogonal basis $\{u_1, u_2, u_3\}$ for W.

Can check $\{b_1, b_2, b_3\}$ L.I.

(Step 1) Let $u_1 = b_1$ and $W_1 = \text{Span}\{b_1\} = \text{Span}\ \underline{\qquad \{u_1\} \qquad}$.

(Step 2) Let $u_2 = b_2 - \text{Proj}_{W_1} b_2$. (Then u_2 is orthogonal with $\underline{\qquad u_1 \qquad}$ (b_1)).

Because $b_2 \notin \text{Span}\{b_1\} = W_1$, ($b_1, b_2$ L.I)

$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = b_1$

$\text{Proj}_{W_1} b_2 = \hat{b}_2$

$(z) \quad b_2 - \hat{b}_2 \in W_1^\perp$

$= y - \hat{y}$

$\neq 0$

$u_2 = b_2 - \text{Proj}_{W_1} b_2$

$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \left(\frac{b_2 \cdot u_1}{u_1 \cdot u_1} \right) \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$

(can scale u_2 to simplify hand calculations)

(Step 3) Let $W_2 = \text{Span}\{u_1, u_2\}$. Then $\text{Proj}_{W_2} b_3 = \frac{b_3 \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{b_3 \cdot u_2}{u_2 \cdot u_2} u_2$.

Let $u_3 = b_3 - \text{Proj}_{W_2} b_3$. Then u_3 is orthogonal with both u_1 and u_2.

Hence, $W = \text{Span}\{u_1, u_2, u_3\}$ where $\{u_1, u_2, u_3\}$ is an orthogonal basis.

And an orthonormal basis for W is $\left\{ \frac{u_1}{\|u_1\|}, \frac{u_2}{\|u_2\|}, \frac{u_3}{\|u_3\|} \right\}$.

4. The Gram-Schmidt Process:

Given a basis $\{b_1, \cdots, b_p\}$ for a subspace W of $\mathbf{R^n}$, define

$$u_1 = b_1$$

$$u_2 = b_2 - \text{Proj}_{W_1} b_2 = b_2 - \frac{b_2 \cdot u_1}{u_1 \cdot u_1} u_1 \qquad W_1 = \text{span}\{u_1\}$$

$$u_3 = b_3 - \text{Proj}_{W_2} b_3 = b_3 - \left[\frac{b_3 \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{b_3 \cdot u_2}{u_2 \cdot u_2} u_2 \right] \qquad W_2 = \text{span}\{u_1, u_2\}$$

$$\vdots$$

$$u_p = b_p - \sum_{i=1}^{p-1} \frac{b_p \cdot u_i}{u_i \cdot u_i} u_i$$

Then $\{u_1, \cdots, u_p\}$ is an orthogonal basis for W.

5. Fill in the blank: Any nonzero subspace W of $\mathbf{R^n}$ has an orthogonal basis.

6. The QR factorization:

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as

$$A = QR$$

where Q is an $m \times n$ matrix whose columns form an orthonormal basis for Col A, and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

Proof: The columns of A form a basis for Col A.

Construct an orthonormal basis $\{u_1, u_2, \cdots, u_n\}$ for $W = \text{Col } A$.

Let $Q = [u_1 \ u_2 \ \cdots \ u_n]$.

For $i = 1, \cdots, n$, $a_i \in \text{Span}\{u_1, u_2, \cdots, u_i\}$. So there are constants such that

$$a_i = r_{1i}u_1 + \cdots + r_{ii}u_i + 0u_{i+1} + \cdots + 0u_n.$$

Can we always choose $r_{ii} > 0$?

Let

$$r_i = \begin{bmatrix} r_{1i} \\ \vdots \\ r_{ii} \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

① Attendance

② Pw. quiz

③ Class #13 #6 ~

④ Quiz # 6 Next Mon. 11/9 (pp #16 ~ #17)

① T/F: $\{u_1, \dots, u_p\}$ orthogonal basis of a subspace W of $\mathbb{R}^n$, $y \in \mathbb{R}^n$

CH T
JS F
BS F

F

$\text{Proj}_W y = UU^T y$

② y in $\mathbb{R}^2$ $\quad y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. $\quad \text{Proj}_W y = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

some nonzero subspace of $\mathbb{R}^2$

The distance from y to W = 1 = $\|y - \hat{y}\|$

③ $W = \text{span}\{u\} \subseteq \mathbb{R}^n$, $u \neq 0$

$y \in W$, $\text{Proj}_W y = \frac{y \cdot u}{u \cdot u} u$ $\quad u \cdot y = c(u \cdot u)$

$c = ?$

④ $A \in \mathbb{R}^{3 \times 5}$

There are 4 subspaces:

Col A, Row A, Nul A, Nul A^T

$\mathbb{R}^3$	$\mathbb{R}^5$
Col A	Row A
Subspaces of Nul A^T	Nul A
$\mathbb{R}^3$ or $\mathbb{R}^5$?	

⑤ GS - process.

W ← subspace

$\{b_1, \dots, b_p\}$ of W

Need basis

$\{u_1, \dots, u_p\}$ orthogonal

Note.

$\text{span}\{b_1, \dots, b_i\}$ = $\text{span}\{u_1, \dots, u_i\}$

$u_1 = b_1$

$u_i^* = b_i^* - \text{Proj}_{W_{i-1}^*} b_i^*$

where $W_{i-1}^* = \text{span}\{u_1, u_2, \dots, u_{i-1}^*\} = \text{span}\{b_1, b_2, \dots, b_{i-1}\}$

$= b_i^* - \sum_{i=1}^{i-1} \left(\frac{b_i \cdot u_i}{u_i \cdot u_i} \right) u_i$

That is, $a_i =$ _____ for $i = 1, \cdots, n$.

Let $R = [r_1 \ r_2 \ \cdots \ r_n]$.

Then, $A = [a_1 \ a_2 \ \cdots \ a_n] = [Qr_1 \ Qr_2 \ \cdots \ Qr_n] = QR$.

Why is R invertible?

Suppose not. $Rx = 0$ has a nontrivial soln $x^* \neq 0$.

$Ax^* = QRx^* = Q0 = 0$

This contradicts A has L.I. columns.

$\therefore$ R should be inv.

diag R > 0 (elements)

$$R = \begin{bmatrix} u_1 \cdot a_1 & u_1 \cdot a_2 & \cdots & u_1 \cdot a_n \\ 0 & u_2 \cdot a_2 & \cdots & u_2 \cdot a_n \\ \vdots & 0 & & \vdots \\ 0 & & & u_n \cdot a_n \end{bmatrix}$$

So, R has _____entries on its diagonal.